

Minimum Time Consensus for a Class of Second Order LTI Systems Under Fuel Constraints

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Abstract:

In this work, a problem of achieving minimum time consensus among a set of N agents modeled as a second order LTI system with bounded inputs and fuel constraints is considered. Unlike our recent work (Rautela et al., 2025), here we consider damping effect in the agent dynamics. First, attainable set for each agent with fuel budget constraints is characterized and its boundary equations are derived. Then, using convexity property, the minimum time at which attainable sets of all agents have non-empty intersection is computed. The computation is made tractable by making use of Helly's theorem. Finally, a procedure is developed to compute minimum time to consensus and the corresponding consensus point for N agents. On application of Helly's theorem, the computation requires finding minimum time to consensus and the corresponding consensus point for each of the NC_3 triplets separately.

Keywords: Linear systems, constrained control, fuel optimal control, multi-agent systems.

1. PROBLEM FORMULATION

Consider a set of N agents described by the second order system

$$\dot{\mathbf{x}}_i = A\mathbf{x}_i + Bu_i, \quad \mathbf{x}_i(0) = \mathbf{x}_{i0} \quad \text{for } i = 1, \dots, N \quad (1)$$

In comparison to pure double integrator dynamics, this model includes the damping. We assume without loss of generality, $A = \begin{bmatrix} 0 & 0 \\ 0 & -b \end{bmatrix}$, $B = \begin{bmatrix} 1/b \\ -1/b \end{bmatrix}$. The system states are $\mathbf{x}_i = [x_{i1} \ x_{i2}]^\top \in \mathbb{R}^2$. The control input to i^{th} agent satisfies the constraint $|u_i(t)| \leq 1$ for all $t \geq 0$. The fuel consumed by i^{th} agent at time t_f is $F(u_i(t), t_f) = \int_0^{t_f} |u_i(t)| dt$. Let the maximum allowable fuel consumption be given as β ; then the *fuel budget constraint* is written as $F(u_i(t), t_f) \leq \beta$.

Definition 1. A multi-agent system is said to achieve *consensus* if for all $i, j \in \{1, \dots, N\}$, where $i \neq j$, $\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| \rightarrow 0$ as $t \rightarrow \bar{t}_f > 0$ and $\mathbf{x}_i(t) = \mathbf{x}_j(t)$ for all $t \geq \bar{t}_f$. The time $\bar{t}_f$ is called the time to consensus, and the point $\mathbf{x}_i(\bar{t}_f) = \bar{\mathbf{x}}$ for $i = 1, \dots, N$ is the corresponding consensus point. The consensus is said to be achieved in finite time if $\bar{t}_f < \infty$.

Problem. Consider a set of N agent system given by (1). Find the minimum time to consensus $\bar{t}_f$ and the corresponding consensus point $\bar{\mathbf{x}} \in \mathbb{R}^2$ such that $\mathbf{x}_i(\bar{t}_f) = \mathbf{x}_j(\bar{t}_f) = \bar{\mathbf{x}}$ and $F(u_i(t), \bar{t}_f) \leq \beta$ for all $i, j \in \{1, 2, \dots, N\}$ with $i \neq j$.

The fuel optimal control $u_i(t)$ profile for i^{th} agent is of bang-off-bang nature (Athans and Falb, 2013).

$$u_i(t) = \begin{cases} \gamma_1, & t \in [0, t_1] \\ 0, & t \in [t_1, t_2] \\ \gamma_2, & t \in [t_2, t_f] \end{cases} \quad 0 \leq t_1 \leq t_2 \leq t_f < \infty \quad (2)$$

where $\gamma_1, \gamma_2 \in \{-1, 1\}$ with $\gamma_1\gamma_2 = -1$. This can be written as sequences $s_1 : (1, 0, -1)$, $s_2 : (0, 1)$, $s_3 : (-1, 0, 1)$, and $s_4 : (0, -1)$. Then fuel consumption for sequences s_1 and s_3 up to time t_f is $t_f - t_2 + t_1$. Similarly, for sequences s_2 and s_4 , $t_f - t_2$. Let the set of admissible control inputs be defined as $U = \{u_i(t) \text{ s.t. (2) and } F(u_i(t), t_f) \leq \beta\}$

Definition 2. *Attainable Set* $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0})$ is the set of all the states that an agent can attain from $\mathbf{x}_{i0} \in \mathbb{R}^2$ using admissible control $u_i(t) \in U$ in time $t_f > 0$. $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0}) = \{e^{At_f} \mathbf{x}_{i0} + \int_0^{t_f} e^{A(t_f-\tau)} Bu_i(\tau) d\tau, \forall u_i(t) \in U\}$

For the boundary of the set $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0})$, we need to only consider the constraint $t_f - t_2 + t_1 = \beta$ which signifies the full utilization of the fuel budget. Also note that the set $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0})$ is a convex set (Rautela et al., 2025).

Attainable Set Characterization: The states attained from $\mathbf{x}_{i0}$ at time t_f using the input profile (2) are

$$\mathbf{x}_i(t_f) = e^{At_f} \mathbf{x}_{i0} + \left(\gamma_1 \int_0^{t_1} + \gamma_2 \int_{t_2}^{t_f} \right) e^{A(t_f-\tau)} B d\tau. \quad (3)$$

Further, ensuring $t_f - t_2 + t_1 = \beta$ gives the boundary of the set $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0})$. Expanding the equation (3) for all sequences s_1, s_2, s_3, s_4 , we obtain boundary expressions for $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0})$. For notation purpose, we denote t_1, t_2

as $t_1^{s_p}, t_2^{s_p}$ for sequences $s_p, p = 1, 2, 3, 4$ and drop the subscript i for brevity. Let $q_1 := e^{-bt_f}, w_1 := e^{\frac{b^2}{2}x_1}, l_1 := e^{\frac{b}{2}(\beta - bx_{10})}, l_2 := e^{\frac{-b}{2}(\beta + bx_{10})}$. Then for s_1 the relation between $\mathbf{x}_{i0} = [x_{10}, x_{20}]$, $\mathbf{x} = [x_1, x_2]$, $t_1^{s_1}, t_2^{s_1}, t_f$ and β is as follows.

$$x_2 - q_1 x_{20} + \frac{1}{b^2} q_1 w_1 l_1 - \frac{q_1}{b^2} - \frac{1}{b^2} + \frac{w_1 l_2}{b^2} = 0 \quad (4)$$

Further, we get $t_1^{s_1} = \frac{\beta + b(x_1 - x_{10})}{2}$ and $t_2^{s_1} = t_f + \frac{b(x_1 - x_{10}) - \beta}{2}$. Similarly, for control input s_3 we get $t_1^{s_3} = \frac{\beta - b(x_1 - x_{10})}{2}$, $t_2^{s_3} = t_f - \frac{(\beta + b(x_1 - x_{10}))}{2}$ and

$$w_1 x_2 - w_1 q_1 x_{20} - \frac{q_1}{b^2 l_2} + \frac{q_1 w_1}{b^2} + \frac{w_1}{b^2} - \frac{1}{l_1 b^2} = 0 \quad (5)$$

For control input s_2 , we get

$$x_1 - x_{10} + \frac{\beta}{b} = 0 \quad (6)$$

$$x_2 - q_1 x_{20} - \frac{1}{b^2} + \frac{e^{-b\beta}}{b^2} = 0. \quad (7)$$

Similarly, for control input s_4 , we get

$$x_1 - x_{10} - \frac{\beta}{b} = 0 \quad (8)$$

$$x_2 - q_1 x_{20} + \frac{1}{b^2} - \frac{e^{-b\beta}}{b^2} = 0. \quad (9)$$

The set of points $\mathbf{x}$ that satisfies equation (4) or (5) forms a part of the boundary $\partial \mathcal{A}^\beta(t_f, \mathbf{x}_0)$ for the two control sequences. Let us denote these two parts by the following sets

$$\partial \mathcal{A}^{\beta, s_1} := \{\mathbf{x} \in \mathbb{R}^2 | (4) \text{ with } 0 \leq t_1^{s_1} \leq t_2^{s_1} < \infty\} \quad (10)$$

$$\partial \mathcal{A}^{\beta, s_3} := \{\mathbf{x} \in \mathbb{R}^2 | (5) \text{ with } 0 \leq t_1^{s_3} \leq t_2^{s_3} < \infty\} \quad (11)$$

From equation (7) and equation (9), we see that this part of the boundary has fixed x_1 co-ordinate. Thus,

$$\partial \mathcal{A}^{\beta, s_2} = \{\mathbf{x} \in \mathbb{R}^2 | x_1 - x_{10} + \frac{\beta}{b} = 0, 0 = t_1^{s_2} \leq t_2^{s_2} < \infty\} \quad (12)$$

$$\partial \mathcal{A}^{\beta, s_4} = \{\mathbf{x} \in \mathbb{R}^2 | x_1 - x_{10} - \frac{\beta}{b} = 0, 0 = t_1^{s_4} \leq t_2^{s_4} < \infty\} \quad (13)$$

The boundary of the set is $\mathcal{A}^\beta(t_f, \mathbf{x}_0) = \bigcup_{i=1}^4 \partial \mathcal{A}^{\beta, s_p}$. An example of attainable set is shown in Figure 1a.

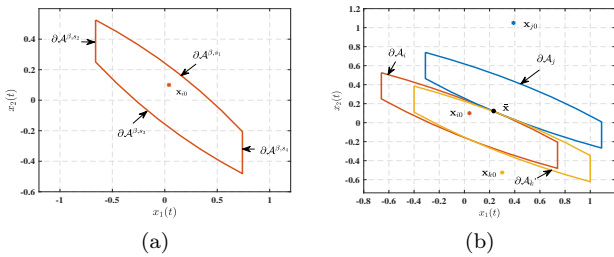


Fig. 1. Attainable set and their intersection

2. MINIMUM TIME CONSENSUS COMPUTATION

Unlike in (Rautela et al., 2025), we note that for fixed fuel budget β , in the (x_1, x_2) -plane the set $\mathcal{A}^\beta(t_f, \mathbf{x}_0)$ converges to a bounded set as $t_f \rightarrow \infty$. Consequently, not all set of initial conditions can attain consensus. We therefore consider only those initial conditions for which consensus is possible. In convex geometry, Helly's theorem (Gruber, 2007), states that for a family of

convex sets in $\mathbb{R}^2$, if every triplet of members in this family has non-empty intersection, then intersection of all members of the family have non-empty intersection. Since $\mathcal{A}^\beta(t_f, \mathbf{x}_0)$ is a convex set in $\mathbb{R}^2$, the problem of computing minimum time consensus point can be broken into computing minimum time to consensus of only a triplet of agents in similar manner as in (Rautela et al., 2025).

Minimum time consensus for $N = 3$: For three agents $\{a_i, a_j, a_k\}$, a time t_{ijk} at which consensus is possible is found by solving $\mathcal{A}_i^\beta(t_{ijk}, \mathbf{x}_{i0}) \cap \mathcal{A}_j^\beta(t_{ijk}, \mathbf{x}_{j0}) \cap \mathcal{A}_k^\beta(t_{ijk}, \mathbf{x}_{k0}) \neq \emptyset$ for t_{ijk} . For minimum t_{ijk} , we need to solve simultaneously equations describing the boundaries of these attainable sets. Therefore, $\bar{t}_{ijk} \geq \max\{\bar{t}_{ij}, \bar{t}_{jk}, \bar{t}_{ik}\}$. Here, $\bar{t}_{ij}$ is the minimum time t_f such that $\mathcal{A}_i^\beta(t_f, \mathbf{x}_{i0}) \cap \mathcal{A}_j^\beta(t_f, \mathbf{x}_{j0}) \neq \emptyset$. Without loss of generality, assume $\bar{t}_{ij} = \max\{\bar{t}_{ij}, \bar{t}_{jk}, \bar{t}_{ik}\}$. We now have two cases: 1) $\bar{\mathbf{x}}_{ij} \in \mathcal{A}_k^\beta(\bar{t}_{ij}, \mathbf{x}_{k0})$ and 2) $\bar{\mathbf{x}}_{ij} \notin \mathcal{A}_k^\beta(\bar{t}_{ij}, \mathbf{x}_{k0})$

For case 1, we just need to compute the solution to the expressions defining the boundary of two attainable sets and $\bar{t}_{ijk} = \bar{t}_{ij}$. In case 2, since at $\bar{t}_{ij}$, the intersection is empty, we need the following condition to be solved for $\bar{t}_{ijk}$: $\partial \mathcal{A}_i^\beta(\bar{t}_{ijk}, \mathbf{x}_{i0}) \cap \partial \mathcal{A}_j^\beta(\bar{t}_{ijk}, \mathbf{x}_{j0}) \cap \partial \mathcal{A}_k^\beta(\bar{t}_{ijk}, \mathbf{x}_{k0}) \neq \emptyset$. Since there are four separate parts to the boundary, to solve these inequality constraints there are 18 cases to be considered. In each of the cases, we solve three equations corresponding to agents a_i, a_j, a_k . The analytical expressions for $\bar{t}_{ijk}$, $\bar{x}_{1,ijk}$ and $\bar{x}_{2,ijk}$ are computed using the Gröbner Basis Cox et al. (2013). On solving three boundary equations we obtain cubic equation in t_{ijk} i.e., $b_3 t_{ijk}^3 + b_2 t_{ijk}^2 + b_1 t_{ijk} + b_0 = 0$.

For a 3 agent case with $\beta \leq 0.7$, $b = 1$, initial conditions $\mathbf{x}_{i0} = [0.01, 0.1]^\top$, $\mathbf{x}_{j0} = [0.39, 1.05]^\top$ and $\mathbf{x}_{k0} = [0.3, -0.525]^\top$, we find $\bar{t}_{ijk} = 1.49$ by solving the equations of degree atmost three for all 18 cases and $\bar{\mathbf{x}}_{ijk} = [0.278, 0.094]^\top$. The intersection of three attainable sets is shown in Fig. 1b.

Minimum time consensus for N agents: From Helly's theorem, it follows that $\bar{t}_f = \max_{1 \leq i, j, k \leq N} \bar{t}_{ijk}$ if and only if attainable set of all ${}^N C_3$ triplets of agents have non-zero intersections at $\bar{t}_f$. The intersection point corresponding to the triplet which has maximum $\bar{t}_{ijk}$ is the minimum time consensus point $\bar{\mathbf{x}}_{ijk}$.

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